# 60A, 300V Ultra Fast Rectifier Rectifier

## FEATURES

- AEC-Q101 qualified available
- Dual rectifier construction, positive center-tap
- Superfast recovery time, high voltage
- Low forward voltage, high current capability
- Low thermal resistance
- Low power loss, high efficiency
- RoHS Compliant
- Halogen-free according to IEC 61249-2-21

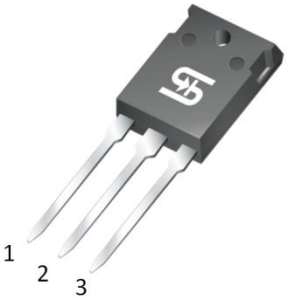
| KEY PARAMETERS | | |
|---|---|---|
| **PARAMETER** | **VALUE** | **UNIT** |
| $I_F$ | 60 | A |
| $V_{RRM}$ | 300 | V |
| $I_{FSM}$ | 300 | A |
| $T_{J\,MAX}$ | 175 | °C |
| Package | TO-247AD (TO-3P) | |
| Configuration | Dual dies | |

## APPLICATIONS

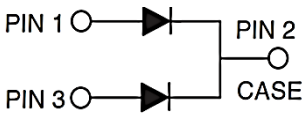
- DC to DC converter
- Switching mode converters and inverters
- Lighting application
- Snubber
- Freewheeling application

## MECHANICAL DATA

- Case: TO-247AD (TO-3P)
- Molding compound meets UL 94V-0 flammability rating
- Terminal: Matte tin plated leads, solderable per J-STD-002
- Meet JESD 201 class 2 whisker test
- Mounting torque: 1.13 N·m maximum
- Polarity: As marked
- Weight: 6.10g (approximately)




**TO-247AD (TO-3P)**




| **ABSOLUTE MAXIMUM RATINGS** ($T_A$ = 25°C unless otherwise noted) | | | |
|---|---|---|---|
| **PARAMETER** | **SYMBOL** | **UG6005PT** | **UNIT** |
| Marking code on the device | | UG6005PT | |
| Repetitive peak reverse voltage | $V_{RRM}$ | 300 | V |
| Reverse voltage, total rms value | $V_{R(RMS)}$ | 210 | V |
| Forward current | $I_F$ | 60 | A |
| Surge peak forward current 8.3ms single half sine wave superimposed on rated load | $I_{FSM}$ | 300 | A |
| Junction temperature | $T_J$ | -55 to +175 | °C |
| Storage temperature | $T_{STG}$ | -55 to +175 | °C |

## THERMAL PERFORMANCE

| PARAMETER | SYMBOL | TYP | UNIT |
|---|---|---|---|
| Junction-to-case thermal resistance | $R_{\Theta JC}$ | 1 | °C/W |

## ELECTRICAL SPECIFICATIONS ($T_A$ = 25°C unless otherwise noted)

| PARAMETER | CONDITIONS | SYMBOL | TYP | MAX | UNIT |
|---|---|---|---|---|---|
| Forward voltage per diode[1] | $I_F$ = 30A, $T_J$ = 25°C | $V_F$ | - | 1.25 | V |
| Reverse current @ rated $V_R$ per diode[2] | $T_J$ = 25°C | $I_R$ | - | 5 | μA |
| | $T_J$ = 125°C | | - | 600 | μA |
| Reverse recovery time | $I_F$ = 0.5A, $I_R$ = 1.0A<br>$I_{rr}$ = 0.25A | $t_{rr}$ | - | 25 | ns |

**Notes:**

1. Pulse test with PW = 0.3ms
2. Pulse test with PW = 30ms

## ORDERING INFORMATION

| ORDERING CODE[1] | PACKAGE | PACKING |
|---|---|---|
| UG6005PT | TO-247AD (TO-3P) | 30 / Tube |
| UG6005PTH | TO-247AD (TO-3P) | 30 / Tube |

**Notes:**

1. “H” means ACE-Q101 qualified

## CHARACTERISTICS CURVES

($T_A$ = 25°C unless otherwise noted)

**Fig.1 Forward Current Derating Curve**

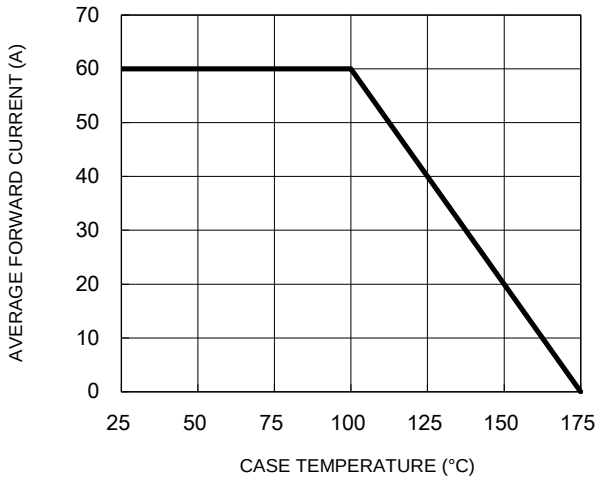



**Fig.2 Typical Junction Capacitance**

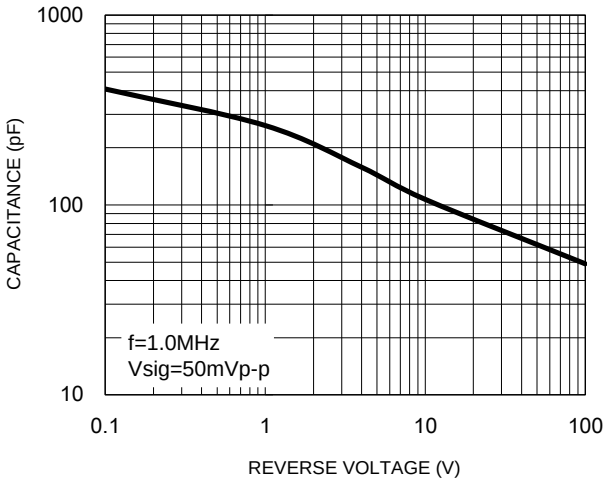



**Fig.3 Typical Reverse Characteristics**

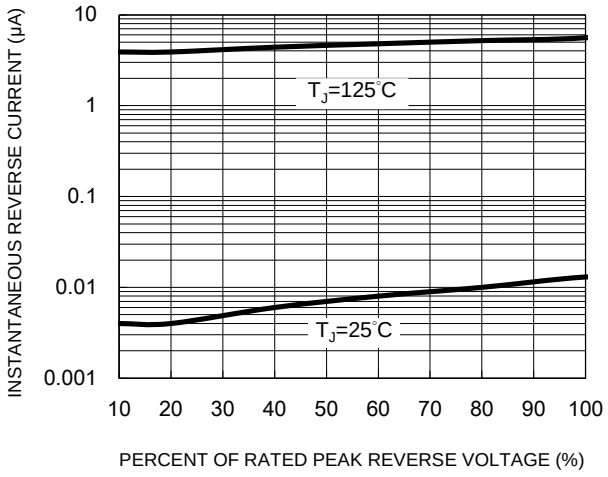



**Fig.4 Typical Forward Characteristics**

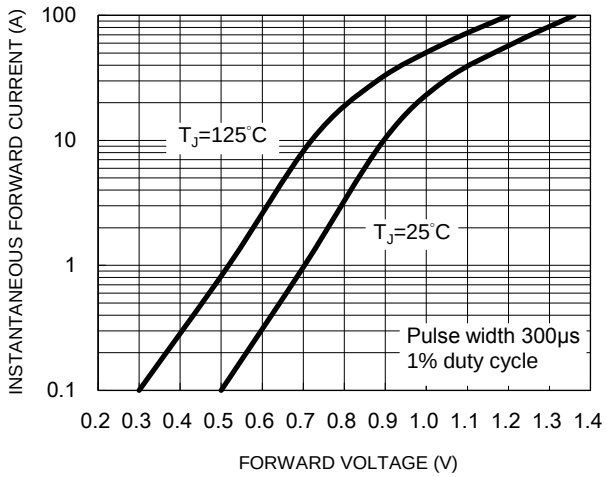
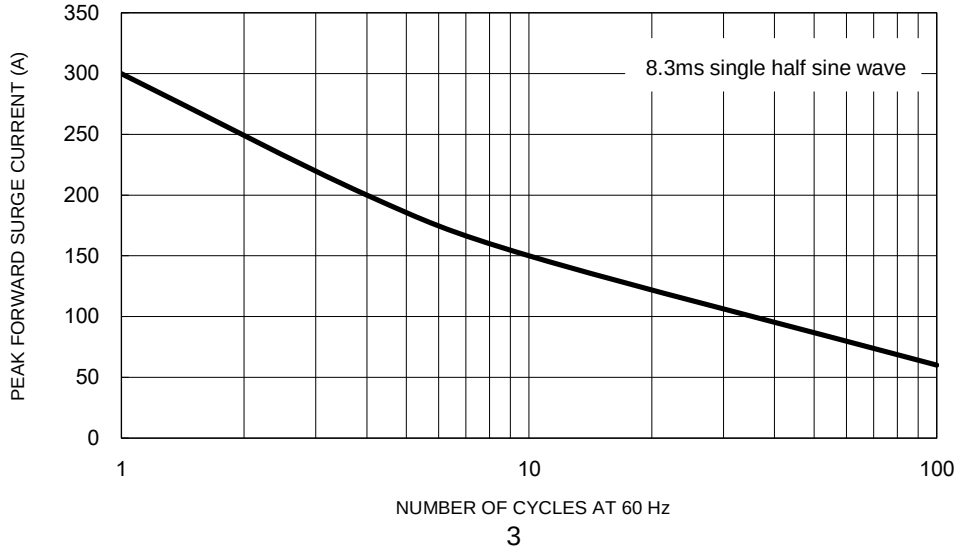


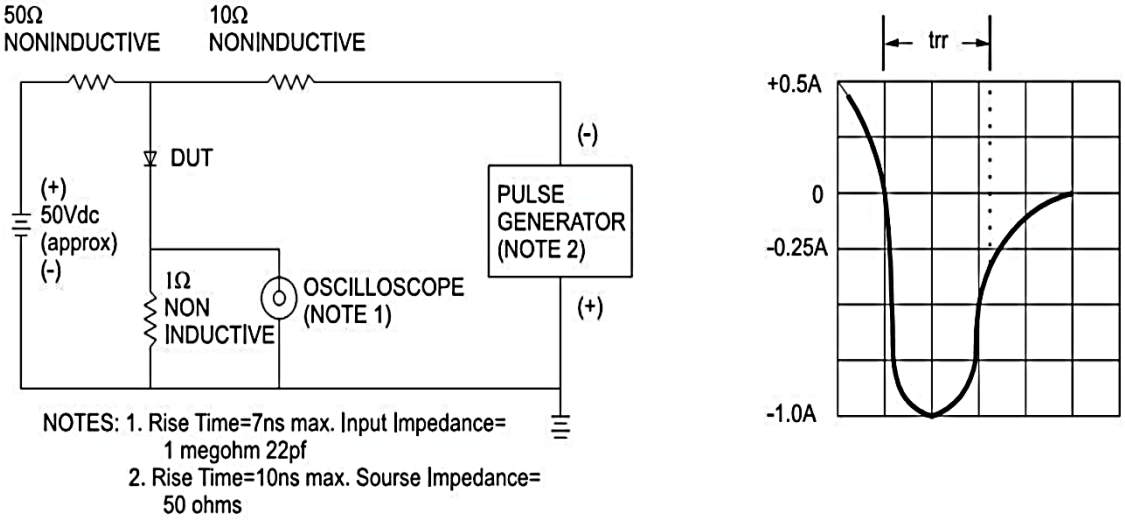

**Fig.5 Maximum Non-Repetitive Forward Surge Current**



## CHARACTERISTICS CURVES

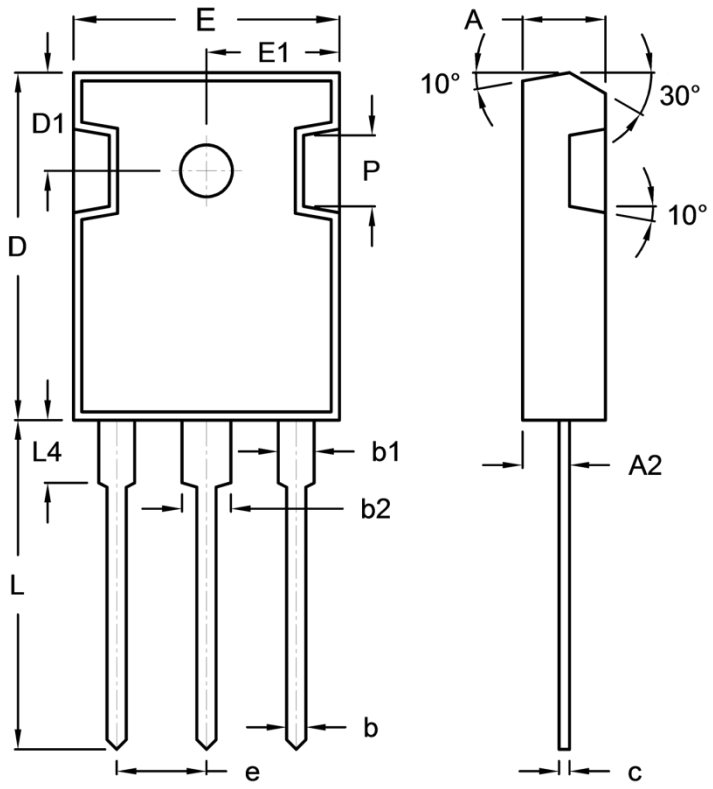
($T_A$ = 25°C unless otherwise noted)

**Fig.6 Reverse Recovery Time Characteristic and Test Circuit Diagram**



## PACKAGE OUTLINE DIMENSIONS

TO-247AD (TO-3P)




| DIM | Unit (mm) | | Unit (inch) | |
|---|---|---|---|---|
| | Min | Max | Min | Max |
| A | 4.90 | 5.16 | 0.193 | 0.203 |
| A2 | 2.70 | 3.00 | 0.106 | 0.118 |
| b | 1.12 | 1.22 | 0.044 | 0.048 |
| b1 | 1.93 | 2.18 | 0.076 | 0.086 |
| b2 | 2.97 | 3.22 | 0.117 | 0.127 |
| c | 0.51 | 0.76 | 0.020 | 0.030 |
| D | 20.80 | 21.30 | 0.819 | 0.839 |
| D1 | 5.70 | 6.20 | 0.224 | 0.244 |
| E | 15.90 | 16.40 | 0.626 | 0.646 |
| E1 | 7.90 | 8.20 | 0.311 | 0.323 |
| e | 5.20 | 5.70 | 0.205 | 0.224 |
| H | 2.90 | 3.40 | 0.114 | 0.134 |
| L | 19.70 | 20.20 | 0.776 | 0.795 |
| L4 | 3.50 | 4.10 | 0.138 | 0.161 |
| P | - | 4.30 | - | 0.169 |

## MARKING DIAGRAM

P/N = Marking Code
G = Green Compound
YWW = Date Code
F = Factory Code

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